Nov 9, 60

High Speed Computational Techniques

Tom Degan 1st Mgr
al Mitchel - (for 1 week)
R. Cypser - now in charge

History:
Feb 5 started
Proposal: - 3 areas Rome - ARPA funds

1 - survey of devices K
2 - prob. analysis B
3 - exploratory design K

* Few people in Bethesda & Kingston *

Meeting after 1 month: Mr. Lilly - ARPA 2 Bosses -- problem
assume: 1000 object cloud
multi-Radar DP system

Dr Wilk, Mr Setter (?) Rome
- want to "do things faster" - ? ---

Took one problem & looked at in depth.

July, meeting with Mr. Lilly
was enthusiastic, Marketing wasn't on board
Semi-annual report: --- was called back same day it was due (Kingston report)
-- renovate in hurry (Bethesda doing most of it)

Rome was very uninterested in detailed study of such probs
wanted general study, -- "SHARE library" type-problems.

- Contract went back to Kingston
More machine org, & devices -- etc
Bethesda - now out except for consulting - Batley part time

There was an internal ARPA document;

BMD stuff, but not multistatic radar systems.

- customer doesn't want to meet with us. - "resubmit original proposal" - sarcastic comment

Oct '60 ; continuation report — was written & submitted ; multistatic DP and other (50 pages worth)

State of art 1968-70 or 1965 period? has continued on — not disturbed, but not much progress,

Bethlehem's part was all done (Multi radar) — removed most of backup in final version.

Kingston hadn't done anything — 10 topics were decided upon.

(- total report wasn't too impressive.

- presented problem to Research re machine organization proposal,

Clayton Andrews, Enerson, L. Johnson, Boehm, Hellerman

Ted Cox - Kingston.

Hierarchy of memories

2D arrays — amount of ordering.

Batley's comment:

Note: more of this type study should be done in Research.

Now about 10 or 12 people in Kingston

R. Cypser, Ted Cox (Honcho)

correlation & classification — Belzhe, Blanche

de ghosting — Cox, Rockefeller, (Batley ½ time)

device survey — C. Jackson, Lampe, B. Hyder

table look-up Techniques: Newton

Lister — Ralph, Bielfeld

Mkt. Mgr: Churchman

High Speed Computational Techniques — NBRB 59-151

ARPA PR #02744 — NBRB 9961

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first meeting before NBRB 7/30/59 — Bid Due Date - 24 Aug 59

Related to "ORDIR"

To Study & Investigate:

- Methods of high density store
- rapid access readout of pre-computed solns for large scale
  - Examples: 1. high density storage - photo film
    2. logical techniques for retrieval of solns,
    3. numerical interpolation techniques,
    4. readout techniques to minimize mem random access time
    5. Methods of precomputing & recording solns.
    6. An estimate of computational equip. to do same job
  - can do breadboard exps.

12 to 14 man years — 1-year contract

Nov. 10 Pine called NPC & wants to know if Res should help.

Following Res. people were originally listed
J. Cocke.
Gil King.
H. E. Petersen

Call to Ted Cox Kempton x 6412

Dr appears not interested — (Bartley was there today)
- what role Res. play?
- are helping in deghosting prob.,
prefer to work separately. (?)

Kellerman
Col. Andrews

- Cox + Rockoff not there last Tues.
- multiprog. prog. being done, machine org'l - several with clients
No. of ops / storage reqts.

at Rome: matrix deghosting, Truck cavell, —
Coord Transformation

they didn't detail problems to be done. So IBM studied in general but won't accomplishing much - needed line problem to go over - ORDIR past experience - put together for multi-static Radar
- Rome wasn't happy with direction — relation to ORDIR didn't want any more ORDIR. They felt we had prob. background already

IBM

Memo Slip

TO LWS Harwood – (over)

NAME Ted Cox - Kingston

+ PHONED
RETURNED YOUR CALL
CALLED TO SEE YOU

HAVE GONE TO	WILL RETURN	PLEASE PHONE NO.
		X6412

A LONG DISTANCE CALL

CALL OPERATOR NO. _____ IN _____

ASK FOR _____ TEL. NO. _____

FOR YOUR	INFORMATION	COMMENTS	SIGNATURE	PLEASE	HANDLE	RETURN	FILE

REMARKS: Please call him regarding contract on High-Speed Computational Techniques

DATE	TIME	SIGNED
11/10	10:00	

Form MO6-1289-4

I do not wish to
interpose myself here
unless I can help.
Betty knows where I
am if I can help-

F.

→ - some difficulty in coming up with future problems
(in air defense alone?)

- Kuyper had never done such a problem like this before
not pulled together -- at loose ends, - didn't lead anywhere -
Bethke tried to pull together, -- need a problem to work on
- good defined prob. -- then design machine

- are considering:

general correlation problem - decoy discrimination
- pattern recog,

associative memory approach,
- beginning

Lorde, C.D. Jackson

Stored pre-computed values -- haven't found

John Newton - Nomograms

Seminar - detail, problems to be discussed
~~Tentative~~ - technology - Kuyper, etc,
Nov 30, Dec 1, 2 - Problems - Ferrell, Walston, Kulsky

Kuyper does favor

call to Ed Adams NHQ

need
- Clear statement whether it will come off or not — if FSD can't do. need letter
- an emergency or not?

another contract from ARPA closely related
- not trying to steal FSD construction contract.

Res. completely covered first by letters
- up to now Res consulting only — Res didn't approve of orig. start

- do not believe that by consulting only — can do it
- what time remains

Res would have to "sell short" need to be drastic
taking responsibility & finishing contract.
- still need FSD people
- work with them (+ supply res. people
Ed Adams will be boss.

NPC -
do some calling now,

- Cypser + Adams + D, home meeting?
- gorman —
- "muffled" bubble chamber presentation —
- C. Andrews group —

- go up + let Ed Adams talk to Cypser —
vote on outcome

NPC called Cooley + he knows about,

Review of Reports

1st prog report: Feb 29, 60 : Some memory feasibility, some circuit performance
2nd " " : Mar 31 : consultations with RADC & ARPA; choice of 4 problems
shows considerable thought in coord. transformation prob.
(still following spirit of contract)
3rd Apr 30 : deghosting problem, trig problem -
(signs of bogging down)
4th May 31 : deghosting
5th Main Report July 11
6th July 31 : deghosting - some memory more.

Bitsby Mem Sept 8: .

[redacted]

Talk with Ed Adams

Cypser doesn't agree with other peoples opinion -

not clear cut decision - even when over - middle on. Thru -

haven't done a good job "make something out of nothing"

[redacted]

Batley

a basic survey { a good one hasn't come out of Kingston yet

need over all supervision from Research

There may be an AF investigation ---

IBM **Memo Slip**

TO Dr. Golsky
Rudy Cypser

✓ PHONED
RETURNED YOUR CALL
CALLED TO SEE YOU

ONE TO	WILL RETURN	PLEASE PHONE NO.

G DISTANCE CALL
OPERATOR NO. _____ IN _____

FOR _____ TEL. NO. _____

R UR	INFORMATION	COMMENTS	SIGNATURE	PLEASE	HANDLE	RETURN	FILE

ARKS: Re: High Speed Computational Techniques Proposal. Appears to be a visit to Kingston in offing - would like to know when this will take place.

DATE 11/14 | TIME 10:30 | SIGNED D-

Form MO6-1289-4

[redacted]

Trip to Kingston Re High Speed Comp Tech.
Meeting with Gypser Nov 19, 60

Gypser to Cooley
~~Date~~: Meeting Report: visit to Rome 11/16/60
"I believe a satisfactory completion of this program now hinges primarily upon the dept of coverage we can provide in the Final Report"

Dr. Welsh
Mr Seltzer, Capt. Geo. Horner } - Intel Lab. will take over

Exploratory Medical Org Group - Mohawk

Letter to Rome Nov 16:

- In Past 2 divisions: (1) State of art technology proj, (2) Problem proj.
- has elected to explor. implementation of formulated problems -
  - systems design,
  - memory logic
  - table lookup
  - areas such as intel,

Gypser is ~~[illegible]~~ resp. for coordination of these efforts,

Seltzer visit

- C. Jackson ✓ - Storage Devices.
- D. Weston ✓ - TLO. Nomograph design - debug. weak now?
- A. Gaspe - logic - components - thin films
- Blasky-Schake - ✓ mem.
- Rockefeller-Cox ✓ - computation

Some names of ~~[illegible]~~. There to RC,
re, interpolation
- Willoughby
- Gilchrist
- Greenstadt
- Wahrhaft
- Kochen
- Tanamoto
- Gilmore

E.D. Jackson - Lewis

graph: capacity vs. access time of contractor report.

cores: can build 4X or 16X larger than present 65K memory, about as far as one can go. --

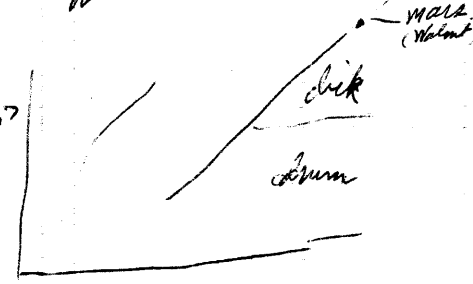
hierarchies:

gap from 20 μs to 1 ms.

cost of cores ~ 10¢ a bit. ($10^{\text{¢}}$-$30^{\text{¢}}$ mfg cost)

" " drums ~ 2¢ per bit

disk 0.1¢ per bit



Read-only: ⇒ capacitor: - much cheaper -- less power

(15 to 1 against to noise)

photodisk --- not as good as disk.

cost per bit is about same only

Reliability -- record not changed. -- can't be change

--- drum disk -- deliberate

core -- change all the time. regen

ϕ's -- need to be changing all the time.

For Report:

will expand on cryogenics and thermoplastics "

plot of thru-put vs capacity (card tape)

" vs. word basis for high speed

Reliability,

Cost -- low adv., production, size, rugged

Read only (Read opt.) conventional core Write only

(R Petersen's concept)

organizations:

associative:

sage control bit "empty bit"

Informer - disk format - metadata descriptors,

addressing:

conceptual

2D approach

scanning - cyclical --

sequential

hierarchies

Info. Retrieval: ~~Radar~~ (applications)

(T. Cox) (60 KMc Esaki diodes at RCA?)

Capt. Homer at Rome - taking over.

- recognized reduction -- no cost to Govt.

Panel Discussions

Will present: 16 sets of data - checking functions. 2 sets of 1M words.

4 valued correlation.

Research in work on } Herb Hellerman,
same problem }

For first Repeat. Will write multivariable correlation -- more general d.p. problem prob. anal. - deghosting refer to this

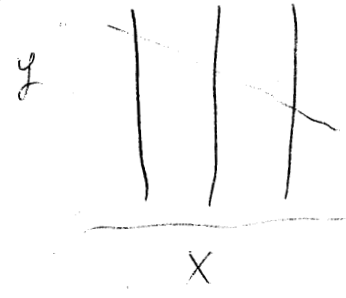
(I. Newton) Precomputed Solns,

Nomography
function tables
Table search
- list, chaining, indexed addr,
arith. opns. (1620 TLO,), 7040 does some,
$\Rightarrow$ interpolation (divided) (not much use)

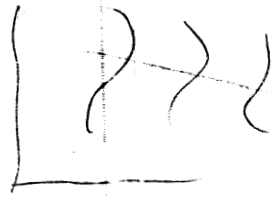
{ Ted Rivlin, L.H. Thomas. ✓
Ralph Willoughby ✓
Don Quarles ✓
Stallberg,
Gilevit
Backman ✓

Carl Kossack
John Griffith

Nomograms:

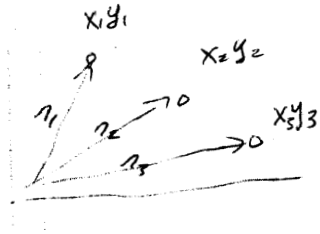


more general - curved lines



3 points on line means this det. is zero $\rightarrow$

$$\begin{vmatrix} X_1 & Y_1 & 1 \\ X_2 & Y_2 & 1 \\ X_3 & Y_3 & 1 \end{vmatrix} = 0$$



In vector terminology:

$$(\bar{r}_3 - \bar{r}_1) \times (\bar{r}_2 - \bar{r}_1) = 0$$

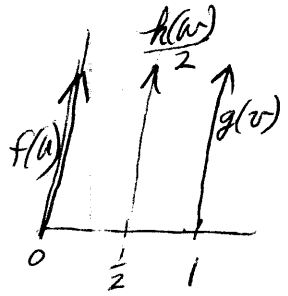
For family of curves $\bar{r}_1 = f_1(u)\,\bar{i} + f_2(u)\,\bar{j}$

$$\begin{vmatrix} f_1(u) & f_2(u) & 1 \\ g_1(v) & g_2(v) & 1 \\ h_1(w) & h_2(w) & 1 \end{vmatrix} = 0$$

can do same in 3 dim.

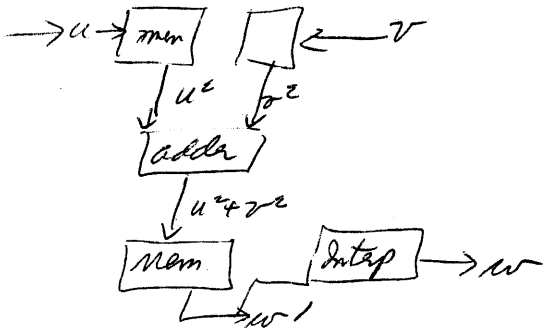
Simple Example:

$$\begin{vmatrix} 0 & f(u) & 1 \\ 1 & g(v) & 1 \\ \frac{1}{2} & \frac{h(w)}{2} & 1 \end{vmatrix} = 0$$



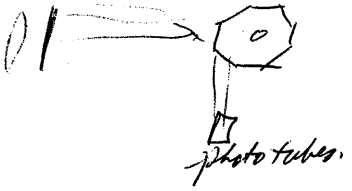
same as $h(w) = f(u) + g(v)$

take case $w^2 = u^2 + v^2$



can store $f(u)$ as table, or by increments, — have to add up

MIT spinning mirror store



list processing { Gelernter, W. Heising, J. Bacus

(AR Plowsky)

- 2 bit multiplier for correlation
- all over counters, etc.

(Cyprus) Follow on - continuous plans
multistatic radar (Bethesda)
(see p20) geophysical technique study (could be Research) — if FSD help is in it too

did design with { C. Andrews said "FSD should
J. Gibson have 50 people"
M. Lesser on such work

Bethesda -- applications oriented
with Res

Cyprus has no strong feeling

→ Cyprus & I visit some of these Research types — get some one to help, to write "term papers" on some of subjects!

- ~~Interpolation~~
- ~~precomputed selections~~
- ~~Indexing~~ Tamamoto

Date: November 18, 1960
From (Dept/Loc): 500/Kingston Eng. Lab.

IBM

Subject and/or Reference: Visit to Rome on High Speed Comp. Tech. Contract
Memo from Contracts to Rome, 11/16/60

To: Mr. H. E. Cooley, Dept. 500, KEL

cc: Mr. L. W. Chuchman, FSD, Utica, N.Y.
Mr. T. C. Cox, Dept. 500, KEL
Mr. N. P. Edwards, FSD, Rockville, Md.
Dr. A. H. Mitchell, Systems Center, Bethesda, Md.
Mr. L. R. Walters, Systems Center, Bethesda, Md.

Messrs. T. C. Cox, L. W. Chuchman and myself visited Mr. M. Setrin and Dr. Wilk to discuss the High Speed Computational Techniques contract. Also attending was Capt. George Horner of the Intelligence Laboratory, who, it appears, will be taking over the responsibility for work of this nature from Mr. Setrin.

I presented a resume of the tasks being pursued and the personnel concerned with these tasks, including the material in the referenced letter on the same subject.

We reviewed the dissatisfaction expressed concerning the Semi-annual Report and the work prior to that time. I made it very clear that by the supplementary activities we have added, we are more than compensating for the over emphasis which Mr. Setrin felt we had placed on the problem definition during the first six months. The work devoted solely to problem definition in that period is estimated to be about 25 man months. (The amount originally planned for applications analysis during the entire contract was about 12 man months.) Through the various supplementary programs, mentioned in the attached letter, the additional effort in the non-program definition areas will amount to 15-20 man months. I made clear the firm conviction of members of our organization that the emphasis on problem definition was in the best interest of the Air Force. I stated that the additional effort being applied to this contract in the non-problem definition areas was initiated so that the program would be completely satisfactory from both points of view.

Mr. Setrin reiterated his early statements that some of the work performed in the study of the multi-static radar systems was redundant with other work being funded by the Air Force. I stated our belief that some of this work appeared to make unique contribution to the data processing feasibility of this type of problem, but that future work along these lines should be conducted only under funding separate from the present High Speed Computational Techniques contract.

In response to a further question, we reiterated the point that the problem definition in the deghosting area had been completely frozen and that the only work in this area being charged to the contract concerned exploratory implementations of the defined problem. The use of separate internal funding to continue the problem definition covering the multiple look approach was explained. (It was clear that Mr. Setrin had not understood the approach pursued and was

11/21

not in a position to evaluate it.)

Mr. Setrin stated that the dissatisfaction with the Semi-annual Report and the work preceding it was historical and there was no value in belaboring it.

Capt. Horner stated the particular interest of the Intelligence Laboratory in the low temperature logic possibilities and stated their intention of working towards an associative memory with IBM under separate contracting.

The remainder of the four hour discussion ranged over various aspects of data processing, including some discussion of cryogenics, Esaki diode developments, parallel processing, table look-up operations, etc. It is clear that Mr. Setrin would prefer to have more emphasis placed on table look-up operations, but he declined to request any shift in emphasis. (Some additional attention will, however, be given to this.)

Previously, Mr. L. W. Chuchman had advised that Mr. Setrin's reaction to the Continuation Statement was "quite favorable as to its content and scope". In this meeting, there was no expression or intimation of dissatisfaction with the current direction or quality of the program. I believe a satisfactory completion of this program now hinges primarily upon the depth of coverage we can provide in the Final Report.

Mr. M. Setrin will visit the Kingston Laboratory on November 22, at which time a more detailed presentation of some of our work and activities in the Lab will be made.

R. J. Cypser

RJC:lms

November 30, 1960

Commander
Rome Air Development Center
[illegible] Air Force Base
Rome, New York

Attention: Mr. Leo Hanlo, RCKL

Subject: Contract AF 30(602)-2161

Dear Sir:

It is the Contractor's desire to keep your agency apprised of the use of additional personnel and resources in the High Speed Computational Techniques study as the later phases of the program are undertaken under the subject contract. This, and a further updating of a statement on our current activities, is the purpose of this note.

As stated previously, the program has been pursued along two complementary directions:

1. Technology projection and problem review involving the following sequence:

   (a) A review of the state of the data processing art in key areas and the prediction of relevant performance characteristics for some future time period.

   (b) A preliminary examination of the utility of combinations of such projected devices in several loosely defined problem classes.

   (c) The consequent partial evaluation of technical [illegible]

2. Problem projection and technology review:

(a) The formulation of a representative problem definition of prime concern to the Air Force, with realizable estimates of values and key parameters.

(b) The exploratory implementation of this solution and the examination of techniques which, in several combinations, take advantage of the unique properties of the problem.

(c) The consequent evaluation of technique combinations in this application.

In the first approach, attention has been given to store units, high-speed logic systems, abbreviation combinations, and, less extensively, to machine organization, special input/output units and mathematical techniques. In the second approach, attention has been given primarily to the decoasting problem in multistatic radars and, to a lesser extent, to tracking, trajectory prediction and coordinate transformation.

With the essential completion of the definition of a projectile decoasting problem emphasis was shifted to the exploratory implementation of the formulated problems and the associated evaluation of relevant techniques. Accordingly, Dr. A. W. Mitchell and Messrs. Gurney and Cantarella of the Systems Center have been primarily in a consulting capacity assisting the implementation studies as discussion requires.

Mr. R. L. Rockefeller, who has had extensive experience in large scale system design problems, has joined Mr. T. G. [illegible] in the exploratory implementation of such systems and the evaluation of the techniques selected. Mr. J. Gurley of the Systems Center, who has also had extensive design experience, has been employed on a part-time basis in this work.

Furthermore, the resources of the Exploratory Machine Organization group of the IBM Research Center at Yorktown have been brought to bear on this problem in a consulting capacity. A small group of the most qualified men in the field at that location have been concerned with guiding the study of alternative implementations and with evolving

at least one preferred combination of techniques.

Mr. J. Spencer has continued studies of the Bible lookup procedures with recent emphasis on monograph techniques.

Mr. A. T. Fleming has concentrated on the examination of memory logic systems with particular emphasis on the possible new functions achievable with cryotron logic. In this, he has been greatly assisted by the work of Mr. F. A. Bolding, who has been providing specific associative memory configurations, under separate internal funding. Valuable assistance has also been obtained from Mr. J. Auslander and his group of the Data Systems Division who, again under separate internal funding, have long been providing subjects of this nature.

The early work on device and technology evaluation and production has been aided and correlated with the thinking of other laboratories within the IBM Corporation, through the assistance of Messrs. A. M. Lampe and C. B. Jackson who have, for some time, been concerned with similar evaluations in their separate staff responsibilities for planning technical programs for the Kingston Laboratory.

In addition, arrangements are being made to further examine the utility and adequacy of the projected high speed computing techniques in other fields, such as intelligence. These additional preliminary examinations will be organized as special inter-Laboratory work sessions. Personnel with detailed knowledge of these areas will participate in consulting capacity.

The IBM Corporation fully appreciates the importance of the studies in which we are engaged under this contract and their possible impact on the effectiveness of solutions to major military problems. As the foundations of this study have been completed, appropriate personnel of the highest caliber in the data processing field have been applied. The resources of three Laboratories of the IBM Corporation are being effectively coordinated in this work. IBM is, accordingly, confident that the basic objectives of this program will be attained.

Dr. N. I. Cypser, Manager of Technical Programs for the Kingston Laboratory, has been made responsible for the coordination of these efforts to insure the fullest possible effectiveness.

Commander -6- November 14, 1940

The breadth of the studies undertaken has necessitated that considerable additional effort beyond the originally planned expenditure be applied. However, the revised spending plan is made [illegible] there will be no overrun on this contract.

If any further information is required, please [illegible] us [illegible] desirable.

Very truly yours,

H. J. Armantrout
Contracts Administrator

HJA:WPS:mt

bc: Dr. H. Kolsky, FSD, Rockville, Md.
Mr. M. Lesser, Research Center